

ABARES Working Paper

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Introduction

Hydrologic models typically predict irrigation water demands via bio-physical processes in-line with FAO-56 standards. However, irrigation demands also depend heavily on the economic behaviour of farmers, particularly their responses to water and crop prices. This study develops a novel method for predicting monthly irrigation water demand that integrates bio-physical processes with an economic profit maximization framework.

This method yields a set of simple parametric equations for predicting annual crop areas and monthly water use as a function of both weather and prices. We apply this method to the Australian Murray-Darling Basin (MDB) with a dataset covering 13 regions and 12 irrigation activities between 2004-05 and 2021-22. Model parameters are obtained using structural estimation, with a joint system of physical and behavioural regression equations solved by non-linear least squares. Validation results show strong performance for water use particularly in the southern basin (annual in-sample R^2 0.94, cross-validated R^2 0.90).

Performance is slightly weaker in the northern basin partly on account of data quality issues (annual in-sample R^2 0.84, cross-validated R^2 0.71). The model is applied to measure the effects on water demand of long-term adjustment in the irrigation sector, including the emergence of almond and cotton crops in the southern basin. The results show that new almond plantings have contributed to a 40 per cent increase in peak summer demands in the lower Murray since 2014.

In future, this bio-economic approach could provide a foundation for integrated hydro-economic models capable of analysing complex water policy issues, including environmental water management, water market design and climate change adaptation.

Source: <https://www.agriculture.gov.au/abares/research-topics/working-papers/bio-economic-approach-predicting-monthly-irrigation-water-demands>